Audit Memo 1: XLRM Framework

CEVE 421/521 Final Project

2026-03-27

1 Overview

Due: Friday of Week 10 (March 27)

This memo maps the complete XLRM decision framework of your chosen climate plan. You will identify the Uncertainties (X), Levers (L), Relationships/System Models (R), and Metrics (M) that define the plan's decision context.

2 Learning Objectives

By completing this memo, you will demonstrate your ability to:

1. Apply the XLRM framework to a real-world climate planning document
2. Extract decision variables, performance measures, system models, and key uncertainties
3. Critique the completeness of a plan's decision framework

3 Requirements

Submit a **3-4 page memo** (PDF) containing the following. Focus on the most important items — aim for 3-5 entries per table rather than an exhaustive list.

3.1 1. Uncertainties (X)

Identify the key uncertainties affecting the plan:

Uncertainty	Type	How Addressed
Climate projections	*Deep (epistemic)*	*Multiple scenarios? Single projection?*
Population growth	*Parametric*	*Sensitivity analysis?*
Damage functions	*Structural*	*Acknowledged?*

Categorize uncertainties as:

- **Parametric:** Known unknowns with estimable ranges
- **Structural:** Model form uncertainty
- **Deep:** Fundamental disagreement or unknown unknowns

3.2 2. Levers (L)

Create a table of the plan's key decision variables:

Lever	Description	Range/Options	Currently Proposed
Example: Seawall height	*Height of coastal protection*	*0-15 ft*	*10 ft*

For each lever, address:

- Is the lever clearly defined with specific options?
- Are alternative options considered, or is only one choice presented?
- What constraints limit the lever's range?

3.3 3. Relationships / System Models (R)

Identify the models used to predict outcomes:

Model Component	Description	Key Assumptions
Hazard model	*e.g., FEMA flood maps*	*Assumes stationarity?*
Exposure model	*e.g., parcel data*	*Current or projected?*
Vulnerability model	*e.g., depth-damage functions*	*Building types?*

For each model, address:

- Are the models appropriate for the hazard and decision context?
- What simplifications or assumptions might affect results?
- Are model limitations acknowledged?

3.4 4. Metrics (M)

Create a table of the plan's performance metrics:

Metric	Description	Units	Target/Threshold
Example: Annual expected damages	*Probabilistic flood losses*	*$/year*	*< $10M*

For each metric, address:

- Is the metric quantifiable and measurable?
- How is the metric calculated or estimated?
- What time horizon does the metric consider?

3.5 5. Framework Critique

In 1-2 paragraphs, assess the overall XLRM framework:

- **Completeness:** Are there important uncertainties, levers, models, or metrics missing?
- **Clarity:** Are the components well-defined and specific?
- **Connections:** Does the plan clearly link uncertainties to models to metrics?

4 Citations

Use proper citations when referencing specific sections, data, or claims from the plan. Include page numbers or section references so the reader can verify your analysis (e.g., "City of Houston, 2020, p. 42"). Use American Geophysical Union (AGU) reference style. You may use any reference management software; we recommend Zotero with the MS Word plugin or BibTeX with LaTeX/Typst.

5 Submission

Submit your memo as a PDF to Canvas by 11:59 PM on the due date.

6 Grading Rubric

Criterion	Points
Uncertainties clearly identified and categorized	15
Levers clearly identified and described	15
System models clearly identified with assumptions	15
Metrics clearly identified and described	15
Thoughtful critique of framework completeness	20
Professional writing and formatting	20
Total	**100**

Bibliography